

When expected returns and volatilities change over time, the optimal short-run weight changes. In equation (4.5), we can put subscript  $t$ s on the means and standard deviations,  $\mu_t$  and  $\sigma_t$ , respectively, of an asset indicating that these are conditional estimates at time  $t$  of expected returns and volatilities over  $t$  to  $t+1$ . The risk-free rate is likely to vary as well (note that the risk-free rate is known at the beginning of the period, so the risk-free rate from  $t$  to  $t+1$  is denoted as  $r_{f,t}$ ). The time-varying short-run weight in equation (4.5) now becomes

$$\text{Short-Run Weight (t)} = \frac{1}{\gamma} \frac{\mu_t - r_{f,t}}{\sigma_t^2}. \quad (4.14)$$

Under time-varying, predictable returns, the optimal long-run strategy comprises the time-varying short-run strategy plus an opportunistic portfolio:

$$\text{Long-Run Weight (t)} = \text{Short-Run Weight (t)} + \text{Opportunistic Weight (t)}. \quad (4.15)$$

The time-varying short-run weight is given in equation (4.14) and is called the *myopic portfolio*. The opportunistic weight is called the *hedging demand* by Merton, who chose the name because the hedging demand portfolio hedges against changes in the investment opportunity set. I prefer to think of it as how the long-run investor can opportunistically take advantage of time-varying returns.<sup>17</sup>

#### *Tactical and Strategic Asset Allocation*

Campbell and Viceira (2002) interpret the Merton–Samuelson portfolios in equation (4.15) as:

$$\begin{aligned} \text{Long-Run Weight (t)} = & \text{Long-Run Myopic Weight} \\ & + [\text{Short-Run Weight (t)} - \text{Long-Run Myopic(t)}] \\ & + \text{Opportunistic Weight (t)} \end{aligned} \quad (4.16)$$

where we split the short-run weight in equation (4.15) into two parts: the average, long-run myopic weight and a deviation from the constant rebalancing weight. We can interpret the first two terms in equation (4.16) as *long-run fixed weights* and *tactical asset allocation*, respectively. The overall long-run weight in equation (4.16) is sometimes called *strategic allocation*.

<sup>17</sup>Equation (4.15) was originally formulated by Samuelson (1969) and Merton (1969, 1971). The opportunistic weight, or hedging demand, for an investor with log utility (CRRA utility with  $\gamma = 1$ ) is zero. Intuitively, a log investor maximizes log returns, and long-horizon log returns are simple sums of one-period returns. Since the portfolio weight is freely chosen each period, the sum is maximized by maximizing each individual term in the sum. That is, a log investor with a long horizon is always a short-run investor.